Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy of MLP by dataset.

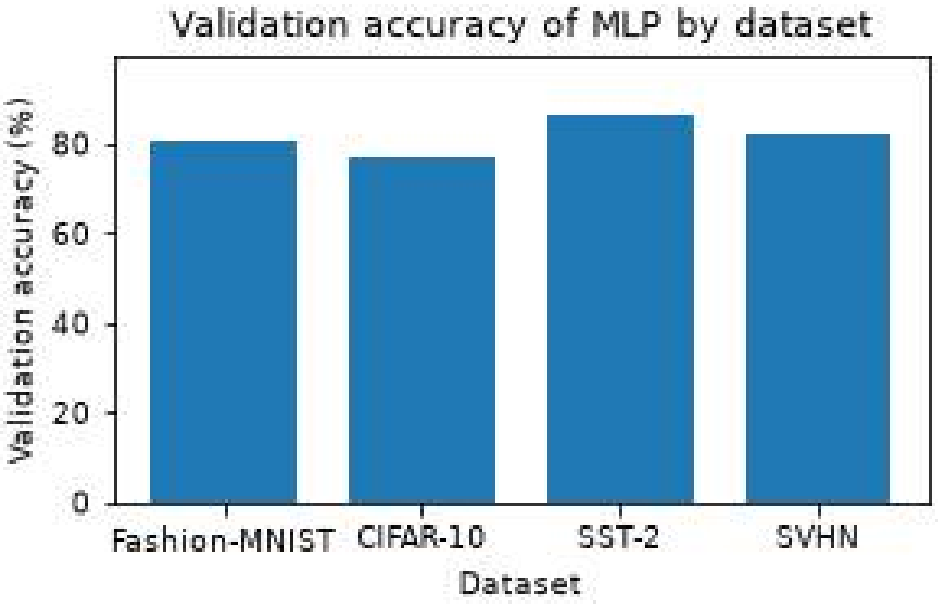


Figure 1: Validation accuracy of MLP by dataset for the models in this experiment.

MLP scores 23.7 points higher on Fashion-MNIST than on CIFAR-10 (Figure 1). MLP achieves 76.7% validation accuracy on CIFAR-10.

Figure 2 shows the validation accuracy by dataset.

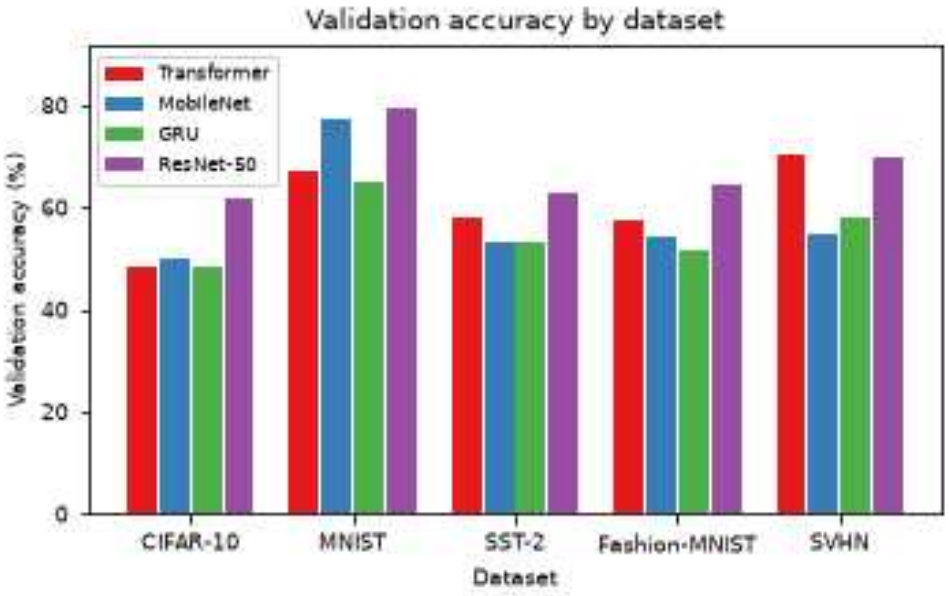


Figure 2: Validation accuracy by dataset for the models in this experiment.

On CIFAR-10, MobileNet obtains a validation accuracy of 95.9%. Transformer achieves 11.6% validation accuracy on Fashion-MNIST.

Figure 3 shows the test accuracy by dataset.

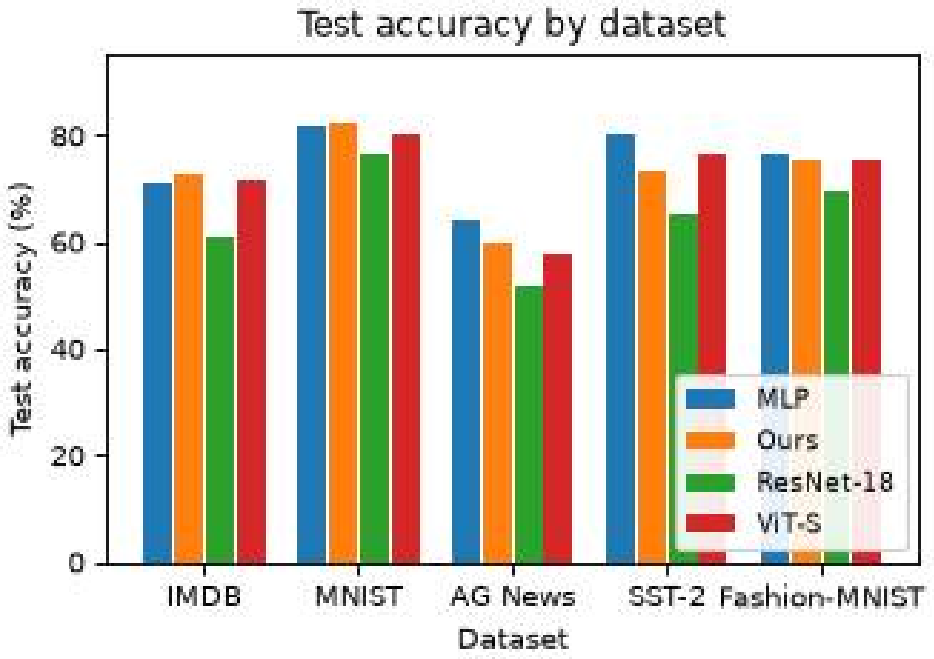


Figure 3: Test accuracy by dataset for the models in this experiment.

As shown in Figure 3, on MNIST, ViT-S is 51.1 points better than ResNet-18. As shown in Figure 3, on AG News, MLP is 22.3 points better than ResNet-18. On IMDB, Ours obtains a test accuracy of 72.6%.

Figure 4 shows the training loss.

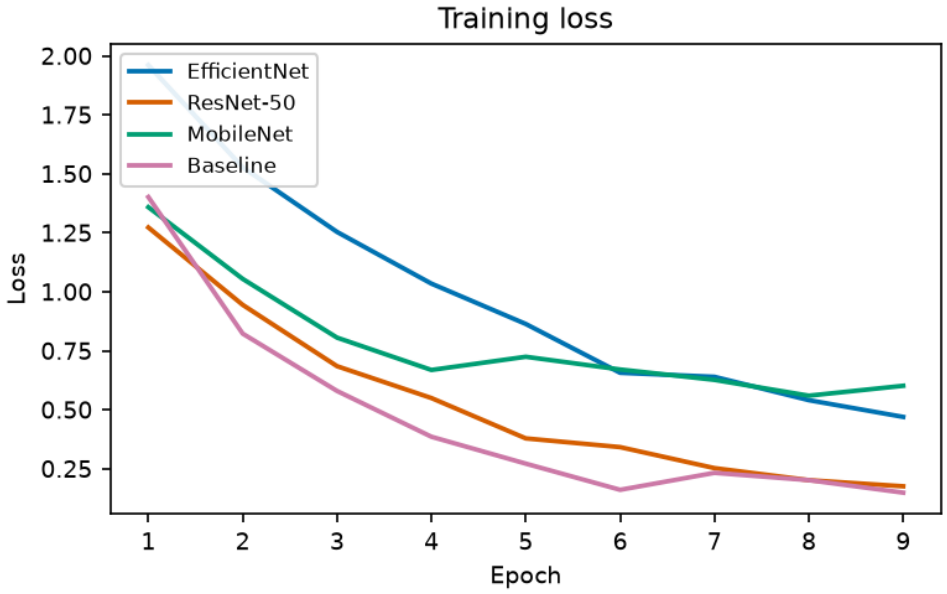


Figure 4: Training loss for the models in this experiment.

At epoch 8, the loss of EfficientNet is 0.541. As shown in Figure 4, at epoch 3, the loss of Baseline is 1.577. As shown in Figure 4, at epoch 1, the loss of EfficientNet is 1.961. As shown in Figure 4, Baseline has a loss of 0.685 at epoch 3.

3 Discussion

On SST-2, SVM obtains an accuracy of 73.0%. The differences between the models are discussed in the next section.